

Chapter 93 — Management of Upper Urinary Tract Obstruction — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Whitaker:** ≤ 12 –15 cm H₂O normal; **>15–20 obstructed**;
perfuse 10 mL/min
- **Differential function <15% (adult)** → kidney non-salvageable
→ nephrectomy
- **Endoscopy fails >2 cm:** stricture or ureteropelvic junction
(UPJ) segment >2 cm → reconstruct
- **Endourology needs $\geq 25\%$ ipsilateral function** (algorithm uses
20%)
- **Pyeloplasty vs endopyelotomy** 10 yr: **75% vs 41%**
- **Crossing vessel:** endopyelotomy **42% vs 86%**; not a
contraindication
- **Endoureterotomy incision:** distal **anteromedial**; proximal
lateral
- **Boari 10–15 cm; psoas hitch +5 cm; descensus 8 cm**
- **Ileal ureter contraindicated:** creatinine **>2 mg/dL**, bladder
dysfunction, inflammatory bowel disease, radiation
- **Retroperitoneal fibrosis (RPF):** L4–L5 aorta; **70% idiopathic**;
prednisone 60 mg → 5 mg

PATIENT PRESENTATION AND DIAGNOSTIC STUDIES — UPJ OBSTRUCTION (UPJO)

- **Commonest neonatal presentation now:** prenatal ultrasound
hydronephrosis (formerly flank mass)
- **Adult presentation:** intermittent flank pain $\pm$ nausea; rarely
hypertension
- **First test neonate:** renal ultrasound; **acute flank pain:** non-
contrast computed tomography (CT)
- **Isotope of choice:** ^{99m}Tc-MAG3 (mercaptoacetyltriglycine)
over DTPA (diethylenetriaminepentaacetic acid)/Hippuran
- **Furosemide 0.3–0.5 mg/kg at 20 min;** obstructed T_{1/2} **>40 min**
- **P40** (tracer at 40 min) raises renogram sensitivity
- **Best post-intervention study:** MAG3 diuretic renogram
- **Whitaker (1973):** invasive → only if non-invasive tests
equivocal; **bladder catheterised**
- **Retrograde pyelogram:** at time of repair (infection risk);
emergently if decompression needed

Indications and options

- **Operate if:** symptoms, ↓ function/progressive ipsilateral loss,
stones, infection, rarely hypertension
- **Neonate operate if:** progressive hydronephrosis or **$\geq 10\%$ fall in
differential function**
- **Gold standard:** dismembered **Anderson–Hynes** pyeloplasty
($\approx 95\%$ success)
- **Primary therapy at major centres:** robotic pyeloplasty
- **Secondary UPJO:** failed endo → pyeloplasty; **failed pyeloplasty**
→ endo
- **Children:** pyeloplasty $\approx$ nephrectomy morbidity <20%;
nephrectomy advised <10%; no agreed cut-off
- **Pre-repair drainage only:** infection, azotaemia (solitary/
bilateral), intractable pain

Endopyelotomy

- **Absolute contraindications:** segment **>2 cm**, active infection,
untreated coagulopathy
- **Incise laterally** (no crossing vessels); high anterior insertion →
true posterior incision
- **Percutaneous access:** mid-posterior/superolateral calyx; **not**
lower pole
- **Percutaneous preferred:** UPJO + stones; marked
hydronephrosis
- **Stones first, then incise** (fragments → fibrosis/granuloma)
- **Stent:** 14/7 Fr (or 10/7, 8 Fr); classic 6 wk; 2 vs 4 wk no
difference
- **Laser:** 200- μ m holmium, **0.8–1.2 J, 10–15 Hz**; 6.9-Fr semirigid
in women

- **Success:** percutaneous 73.5% pooled, 85–90% expert;
retrograde 70–80%
- **Pyeloplasty vs endopyelotomy:** 85/63 (3 yr), 80/55 (5 yr),
75/41 (10 yr)
- **Exclude crossing vessels >4 mm** → retrograde success $\approx 90\%$
- **Balloon dilation alone** decays to $\approx 42\%$; **cautery balloon** →
more bleeding
- **Failed endopyelotomy:** CT angiography; significant vessel →
no repeat endopyelotomy
- **Bleeding:** don't irrigate nephrostomy; **low threshold for**
embolisation
- **Follow-up:** 1 mo, 6 mo, yearly; adults **2–3 yr** (late failures to 36
mo)

Pyeloplasty

- **Stent 4–6 wk;** nephrostomy + stent no benefit in
uncomplicated adult
- **Lap/robotic absolute contraindications:** uncorrected
coagulopathy, untreated urinary tract infection,
cardiopulmonary unfitness
- **Lap success ≥ 90 –95%;** stay 2.7 vs 4.2 d open; conversion 0–
5.5%
- **Robot vs lap:** equal success; advantage = ergonomics
- **Commonest route:** transperitoneal; transmesenteric needs
thin patient, no pre-stent
- **Robotic single-port (SP):** $\approx 50\%$ longer operation; outcomes
similar
- **Drain output ↑ after Foley out** → replace Foley 7 d
- **Late failure:** up to 30% after 2 yr; paediatric 1 in 9 redo

Flaps and alternatives

- **Foley Y-V:** high insertion; **contraindicated** if vessel
transposition needed
- **Culp–DeWeerd spiral:** dependent UPJ + long proximal
narrowing; **length:width $\leq 3:1$**
- **Scardino–Prince vertical:** box-shaped pelvis; shorter flap than
spiral
- **Davis intubated ureterotomy (1943):** long/multiple strictures;
routine nephrostomy
- **Ureterocalicostomy:** small intrarenal pelvis, horseshoe, failed
pyeloplasty; **resect** lower-pole parenchyma
- **Ureterocalicostomy failure:** cortex **<5 mm**, poor function
- **Kuster 1891:** first successful dismembered pyeloplasty
- **Nephrostogram ≥ 7 –10 d;** clamp 12–24 h before removal

RETROCAVAL URETER

- **Persistent posterior cardinal vein; always right**
- **S-shaped** ureter on pyelography; **3D CT** diagnostic
- **Operate if** functional obstruction → divide, transpose anterior
to inferior vena cava; spatulate 1.5–2 cm

URETERAL STRICTURE

- **Exclude malignancy first:** idiopathic → CT; uncertain → ureteroscopy + biopsy/barbotage
- **Stricture after ureteroscopy (URS) ≈3% vs shock-wave lithotripsy 1.5%;** impacted stone >2 mo ≈1 in 4
- **Extrinsic compression:** single stent fails → tandem or metallic stent
- **Metallic stent** changed every 12 months; mesh stents → tumour ingrowth
- **Balloon:** 4 cm × 5–8 mm, until waist gone, 10 min; stent 2–4 wk
- **Balloon best:** iatrogenic non-anastomotic (85%) vs anastomotic (50%); ≤2 cm, ≤3 mo
- **First-line <2 cm:** retrograde holmium endoureterotomy (66–83%)
- **Incision** includes 2–3 mm normal ureter; stent 8–12 Fr; renogram 2 yr
- **Obliterated ureter:** cut to the light; stent 8–10 wk
- **Algorithm:** >2 cm + >20% → repair; <20% → lap nephrectomy

Reconstruction ladder (Table 93.2)

- **Ureteroureterostomy 2–3 cm · reimplant 4–5 · psoas hitch 6–10 · Boari 12–15 (text 10–15) · descensus 5–8**
- **Ureteroureterostomy:** spatulate 5–6 mm at 180°; success >90%
- **Gunshot:** excise margin of normal-looking ureter (blast effect)
- **Leak:** stop suction; Foley + anticholinergic; then nephrostomy
- **Reimplant:** distal 3–4 cm; robotic = open success, shorter stay
- **Psoas hitch:** >85%; contraindicated by small contracted bladder; not above pelvic brim
- **Boari:** commonest complication recurrent stricture; small bladder contraindicates
- **Transureteroureterostomy (TUU) absolute contraindications:** short donor ureter; diseased recipient
- **TUU relative:** stones, RPF, urothelial cancer, chronic pyelonephritis, radiation
- **TUU:** voiding cystogram; tunnel **above inferior mesenteric artery (IMA)** (kinking)
- **Ileal ureter:** ≥15 cm from ileocaecal valve; isoperistaltic; >15 cm → no reflux
- **Ileal ureter creatinine >2** → ≈half hyperchloraemic acidosis
- **Ileal ureter surveillance** endoscopy from year 3
- **Autotransplant:** absent/poor contralateral kidney or no other option (Hardy 1963)
- **Buccal onlay** + omental wrap: scarred ureter; CORRUS series

URETEROENTERIC STRICTURE

- **Conduit 4–9%; continent 3–25%;** mostly <2 yr; **left > right**
- **Cause:** ischaemia (adventitial stripping); left under sigmoid mesentery/IMA
- **Non-refluxing ≈13% vs refluxing ≈2–3%** → use refluxing in low-pressure reservoir
- **Wallace** (joined) 0% vs **Bricker** (separate) 3.7%

- **Chronic conduit hydronephrosis often non-obstructive;** renogram if function falls or reflux lost
- **Gold standard:** open revision (≈91% vs endo ≈26%)
- **Endoscopic:** antegrade (nephrostomy present); internal-external stent (mucus)
- **Endo by length:** <1 cm ≈50% vs >1 cm 6%; **right 41% vs left 19%**
- **Left stricture** → formal repair (sigmoid mesentery bleeding)
- **Laser endoureterotomy** 50–80%; balloon long-term ≤30%

RETROPERITONEAL FIBROSIS

- **Age 40–60; M:F 2–3:1;** incidence 1 per 200,000–500,000
- **Idiopathic 70%** = chronic periaortitis; antigen ceroid
- **Commonest drug:** methysergide/ergots; also β-blockers, phenacetin
- **Commonest malignancy:** lymphoma
- **Imaging:** medial deviation of mid-ureters; smooth taper; usually bilateral
- **Magnetic resonance:** low T1; **high T2 = active**
- **Positron emission tomography:** most sensitive for activity; negative predictive value 100% → may avoid biopsy
- **Uraemic, anuric, critically ill** → nephrostomy; watch post-obstructive diuresis
- **First-line:** steroids, ≈80% response; relapse on taper 25–50%
- **Second-line:** azathioprine, cyclophosphamide, mycophenolate, tamoxifen, rituximab
- **Ureterolysis:** bilateral; start distal; “split and roll”; deep biopsies
- **Protect ureter:** intraperitonealise ≈ lateralise; omental wrap most definitive
- **Failed ureterolysis:** autotransplant; dead kidney → nephrectomy

CLASSIC TRAPS

- **Trap:** dilation = obstruction — **Actually:** needs renogram/Whitaker
- **Trap:** crossing vessel contraindicates endopyelotomy — **Actually:** prognostic only; entanglement matters
- **Trap:** lower-pole access for endopyelotomy — **Actually:** mid-posterior/superolateral calyx
- **Trap:** incise before stone removal — **Actually:** stones first
- **Trap:** irrigate a bleeding nephrostomy — **Actually:** tamponade, then embolise
- **Trap:** simple nephrotomy for ureterocalicostomy — **Actually:** resect parenchyma
- **Trap:** retrocaval ureter on the left — **Actually:** always right
- **Trap:** non-refluxing ureteroenteric protects kidney — **Actually:** strictures more; use refluxing
- **Trap:** endoscopic incision of left ureteroenteric stricture — **Actually:** formal repair
- **Trap:** unilateral ureterolysis for unilateral hydronephrosis — **Actually:** bilateral